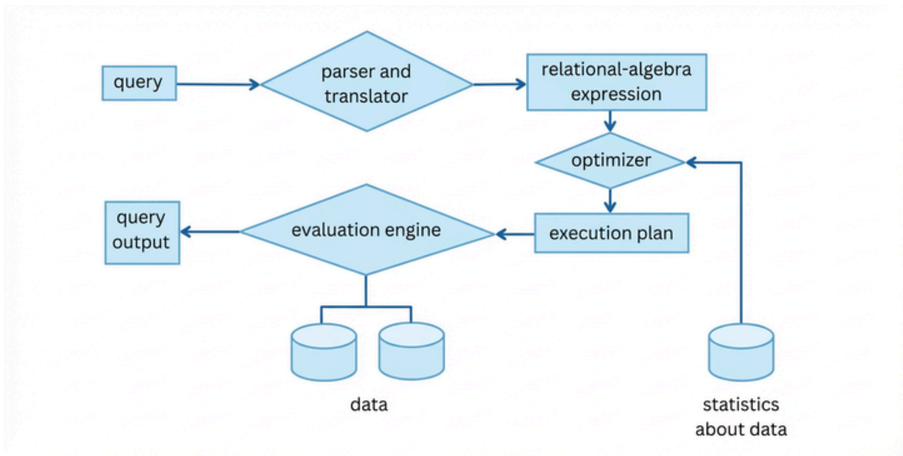
Advanced Database

Dr. M. Ali Nematbakhsh

Project5: Query Processing + Query Optimization + Transaction

Question 1: Explain the query processing process by considering a specific query of your choice. Your explanations must be complete and consistent with the Figure below. Clearly explain each part.



Question 2: Consider the bank Database below. where the Primary Keys are underlined.

branch(*branch_name*, *branch_city*, *assets*)
customer (*customer_name*, *customer_street*, *customer_city*)
loan (*loan_number*, *branch_name*, *amount*)
borrower (*customer_name*, *loan_number*)
account (*account_number*, *branch_name*, *balance*)
depositor (*customer_name*, *account_number*)

This is The Query that you have :

select *T.branch_name*
from *branch T*, *branch S*
where *T.assets* > *S.assets* **and** *S.branch_city* = "Brooklyn"

Write an effictive relational-algebra expression that is equivalent to this query. Justify your choice.

Question 3: Suppose you need to sort a relation of 40 gigabytes, with 4-kilobyte blocks, using a memory size of 40 megabytes. Suppose the cost of a seek is 5 milliseconds, while the disk transfer rate is 40 megabytes per second. Please provide a Detailed Solution.
Find the cost of sorting the relation, in seconds, with b = 1 and with b = 100.

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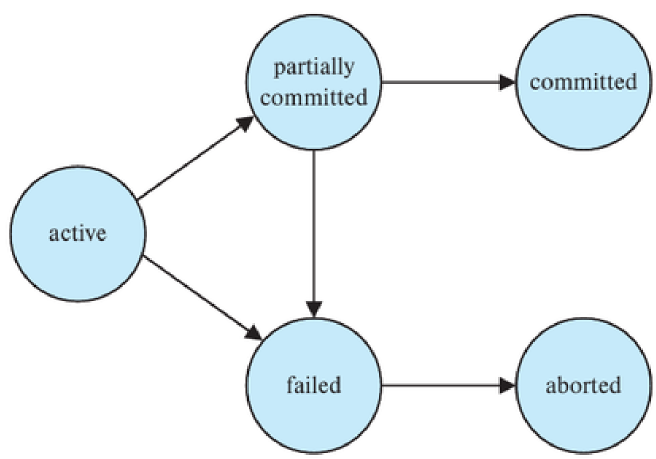
Question 4: **Show that the Following equivalent holds. Please explain how you can apply the, to improve the efficiency of certain queries.**

$$E_1 \bowtie_\theta (E_2 - E_3) \equiv (E_1 \bowtie_\theta E_2 - E_1 \bowtie_\theta E_3).$$

Question 5: **Consider the relations r_1 (A, B, C), r_2(C, D, E), and r3(E, F), with primary keys A, C, and E, respectively. Assume that r_1 has 1000 tuples, r_2 has 1500 tuples, and r3 has 750 tuples. Estimate the size of $r_1 \bowtie r_2 \bowtie r3$, and give an efficient strategy for computing the join.**

Question 6: **Explain the ACID Properties in Database systems. For each of the four components :**
1: Define what it means.
2: Explain why it's important.
3: Provide a simple Example.

Question 7: **A transaction passes through several states during its execution, eventually reaching either commit or abort. The possible states of a transaction are shown in the Diagram below. Describe and explain all possible sequences of a transaction's states.**



Question 8: **Since every conflict-serializable schedule is view-serializable, why do we emphasize conflict-serializability rather than view-serializability?**

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Question 9: **Consider the following two Transactions:**
with A = B = 0 as initial values.
1: Show a concurrent executuion of T13 and T14 that produces a nonserilazable schedule.
2: Is there a concurrent execution of T13 and T14 that produces a serializable Schedule?

T_{13}: read(A);
read(B);
if $A = 0$ **then** $B := B + 1$;
write(B).

T_{14}: read(B);
read(A);
if $B = 0$ **then** $A := A + 1$;
write(A).

Question 10: **Consider the following partial scheduel involving Transactions T10 and T11 and :**

$$S : R_{10}(A), R_{11}(B), W_{10}(A), R_{12}(A), W_{12}(C), R_{11}(A)$$

if Transaction 10 Fails and aborts immediately after the last instruction:
1: Explain exactly what must happen to transactions T12 and T11?
2: Identify the specific term for this phenomenon and briefly explain why it is considerable in databse systems.

What to upload for us?

- Complete answers in one PDF file (It can be handwritten, etc)

All your files must be in one ZIP file named:
DBHW{Project Number}-{Student Complete name}-{Student Number}

Students must follow the Template structure, or they will receive a grade deduction.

Dead Line : 20 January - 30 Day

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